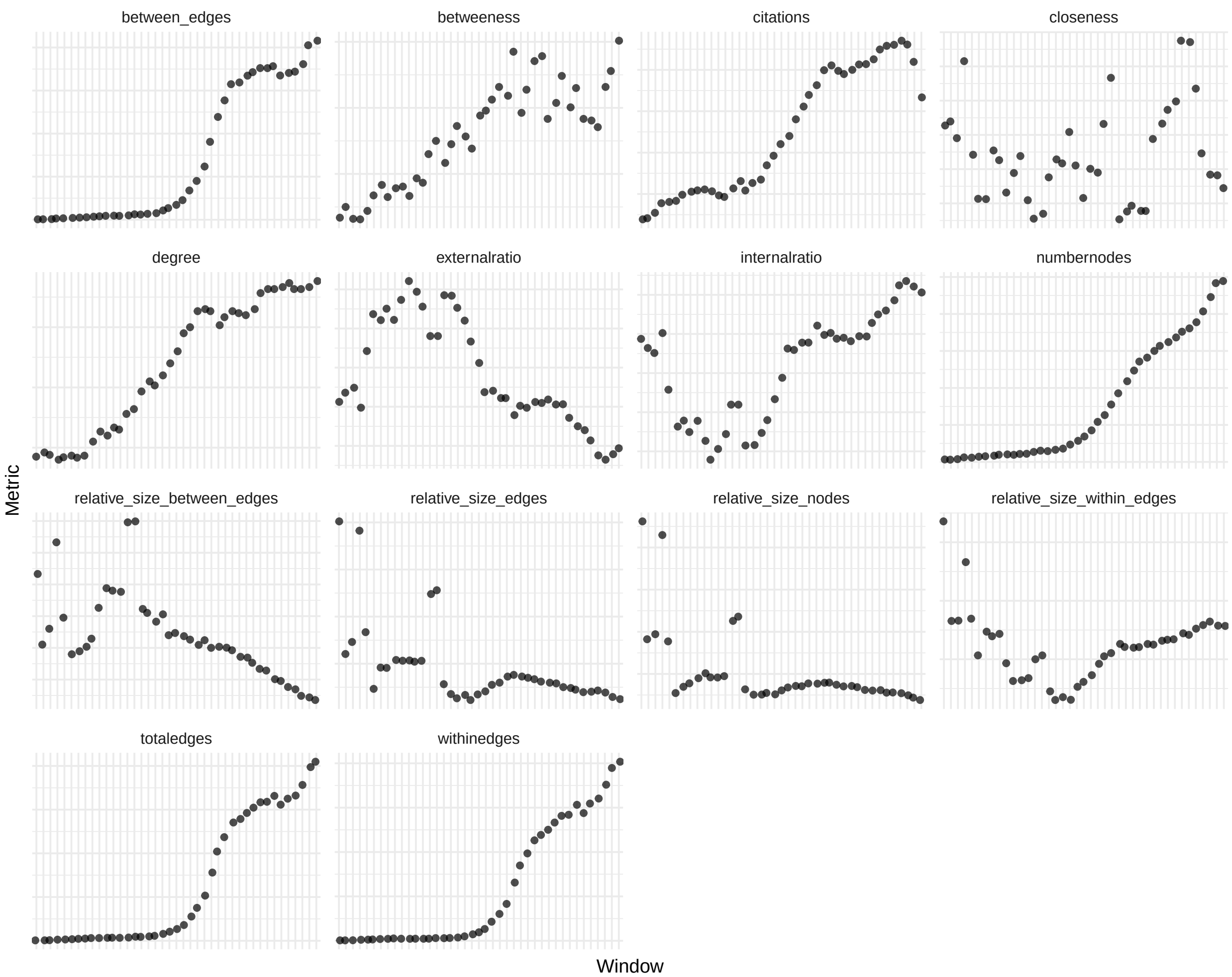
Field-Level Analysis

2025-09-09

1. Aggregate Metrics

The Metrics used here are formally defined on a per-field basis below. The aggregate metrics are computed similarly, but over all fields instead of per field.

1.1 Aggregate Evolution of Metrics



1.2 Number of Nodes and Edges over Time

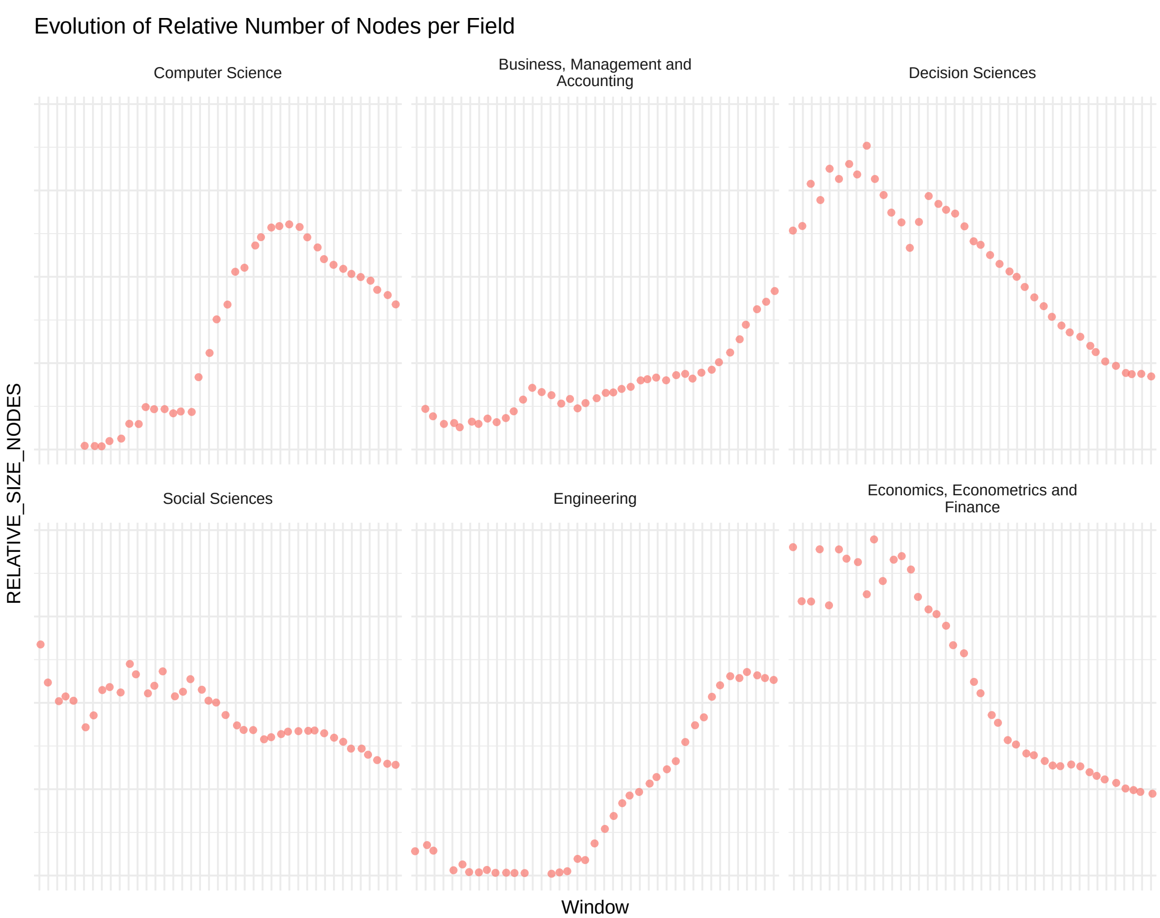
Window	Number of Nodes	Number of Edges
1980-1984	71	203
1981-1985	85	301
1982-1986	104	377
1983-1987	135	582
1984-1988	163	842
1985-1989	233	1182
1986-1990	248	1385
1987-1991	270	1579
1988-1992	307	1900
1989-1993	316	1944
1990-1994	302	2083
1991-1995	339	2183
1992-1996	346	2085
1993-1997	364	1940
1994-1998	406	2433
1995-1999	477	2940
1996-2000	545	3390
1997-2001	598	3879
1998-2002	776	5356
1999-2003	948	7176
2000-2004	1128	9121
2001-2005	1388	12127
2002-2006	1754	18350
2003-2007	2052	24776
2004-2008	2493	33425
2005-2009	2985	50247
2006-2010	3501	66059
2007-2011	3962	76943
2008-2012	4380	88183
2009-2013	4575	91423
2010-2014	4863	96227
2011-2015	5102	100431
2012-2016	5316	105586
2013-2017	5531	106235
2014-2018	5778	111244
2015-2019	5974	105257
2016-2020	6244	109745
2017-2021	6746	111773
2018-2022	7436	120666
2019-2023	8124	136208
2020-2024	8282	141309

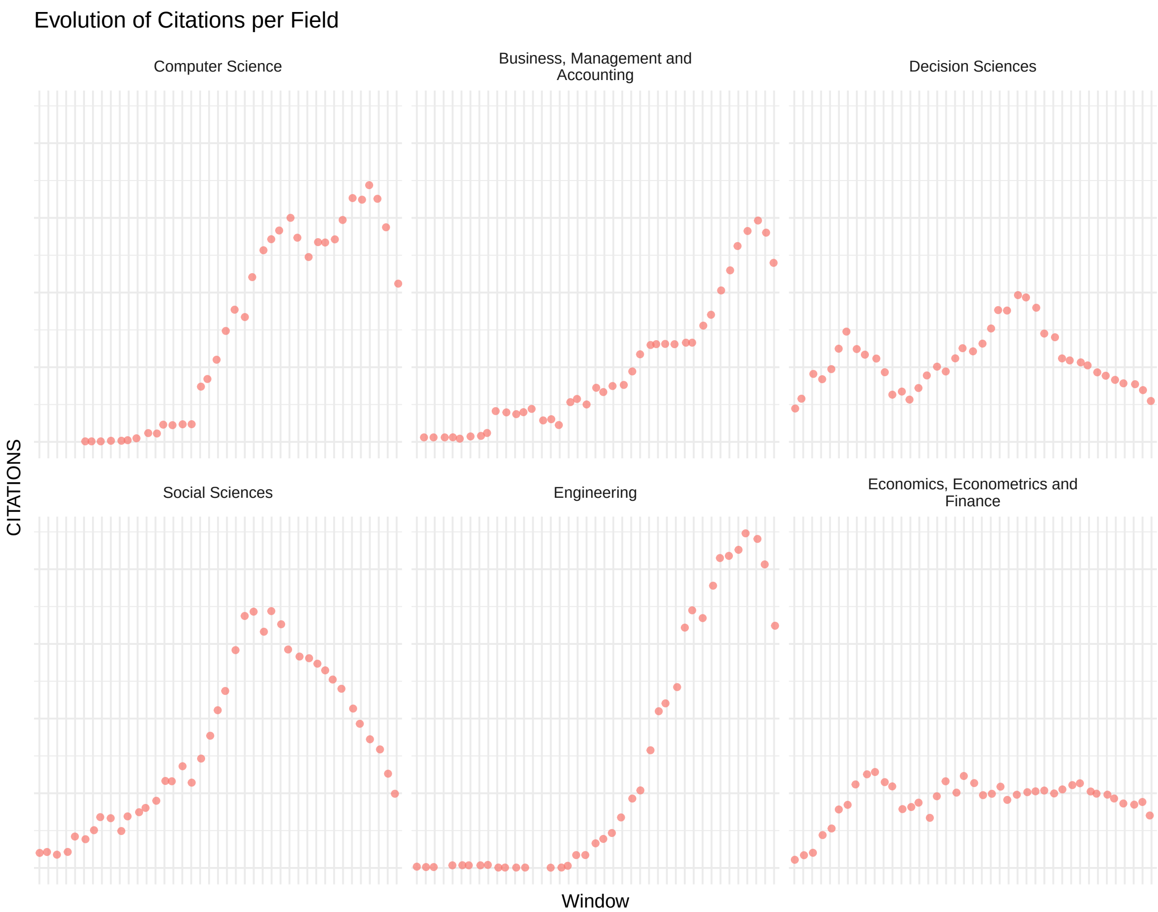
2. Evolution of Fields

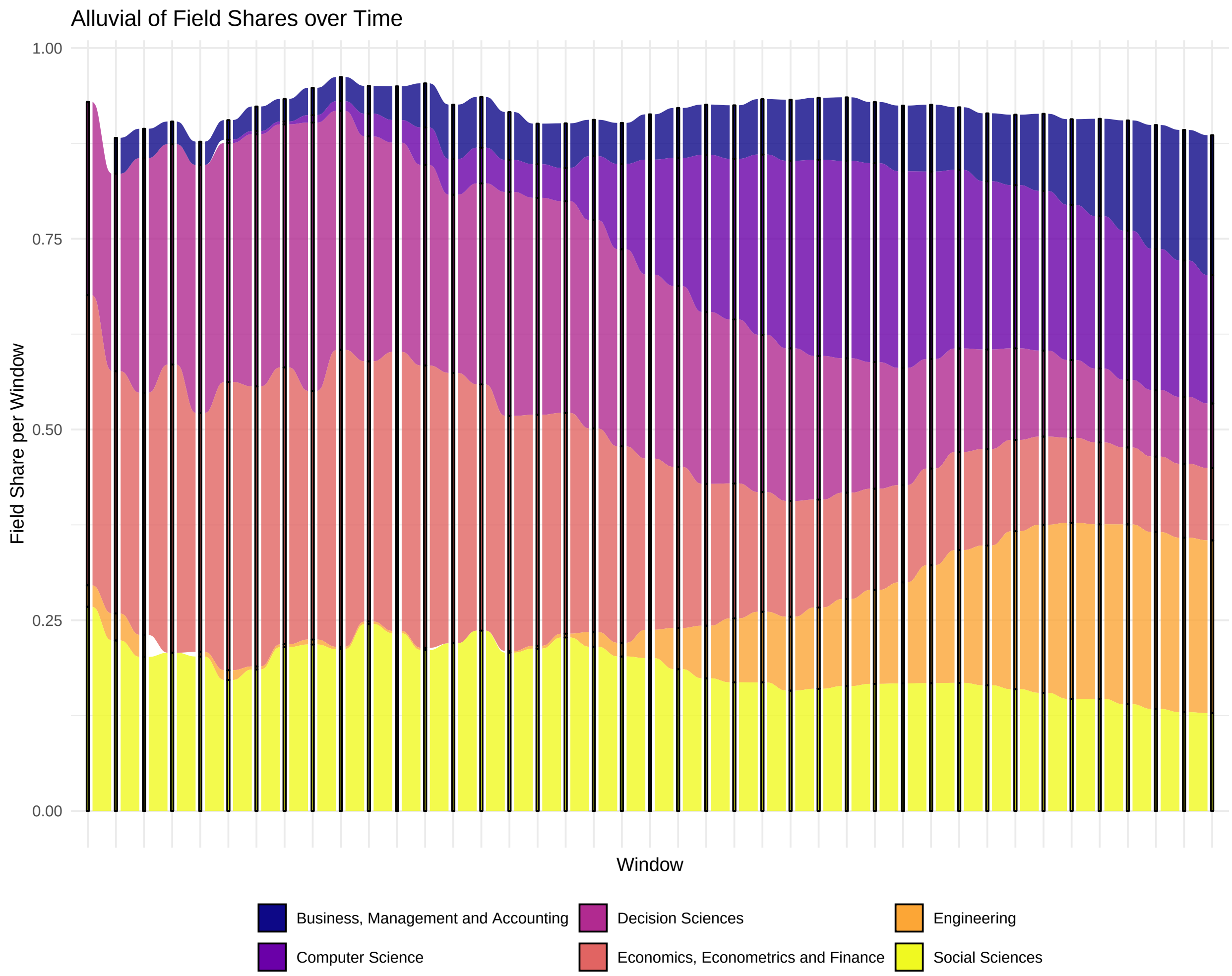
Field-Level Metrics are derived from networks where articles are nodes and edges are based on the strength of bibliographic coupling between articles. The nodes, edges and their information is then aggregated per discipline, here called field.

2. Relative Number of Nodes and Number of Citations

- Relative number of Nodes:
$$RelNodes_{t,f} = \frac{N(V)_{t,f}}{\sum_f^F N(V)_{t,f}} \tag{1}$$
  where:
- $RelNodes_{t,f}$ measures the relative number of nodes for field f in time window t across all article nodes
- $N(V)_{t,f}$ measures the total number of nodes for field f in time window t across all article nodes. It is therefore defined as $N(V)_{t,f} = \sum_{v_i} 1_{field(v_i)=f}$, where $1_{field(v_i)=f}$ is an indicator function, taking the value one if article node v_i belongs to field f.
- The number of citations per field and time window is measured as the sum of citations across all articles of a field in a give time window, respectively.




Evolution of Citations per Field
Computer Science
Business, Management and Accounting
Decision Sciences
Social Sciences
Engineering
Economics, Econometrics and Finance
CITATIONS
Window


Alluvial of Field Shares over Time
1.00
0.75
0.50
0.25
0.00
Field Share per Window
Window
Business, Management and Accounting
Computer Science
Decision Sciences
Economics, Econometrics and Finance
Engineering
Social Sciences

2.2 Relative Number of Edges

- Relative Number of Total Edges:

$$RelEdges_{t,f} = \frac{N(E)_{t,f}}{\sum_f^F N(E)_{t,f}} \tag{2}$$

where:

- $RelEdges_{t,f}$ measures the relative number of edges incident on nodes within field f in time window t
- $N(E)_{t,f}$ measures the total number of edges incident on nodes within field f in time window t. It is therefore defined as $N(E)_{t,f} = \sum_{e=(v_i,v_j)} 1_{field(v_i)=f \vee field(v_j)=f}$, where $1_{field(v_k)=f}$ is an indicator function, taking the value one if article node k belongs to field f
- Relative Number of Within-Field Edges:

$$RelEdgesWithin_{t,f} = \frac{N(E_{within})_{t,f}}{\sum_f^F N(E)_{t,f}} \tag{3}$$

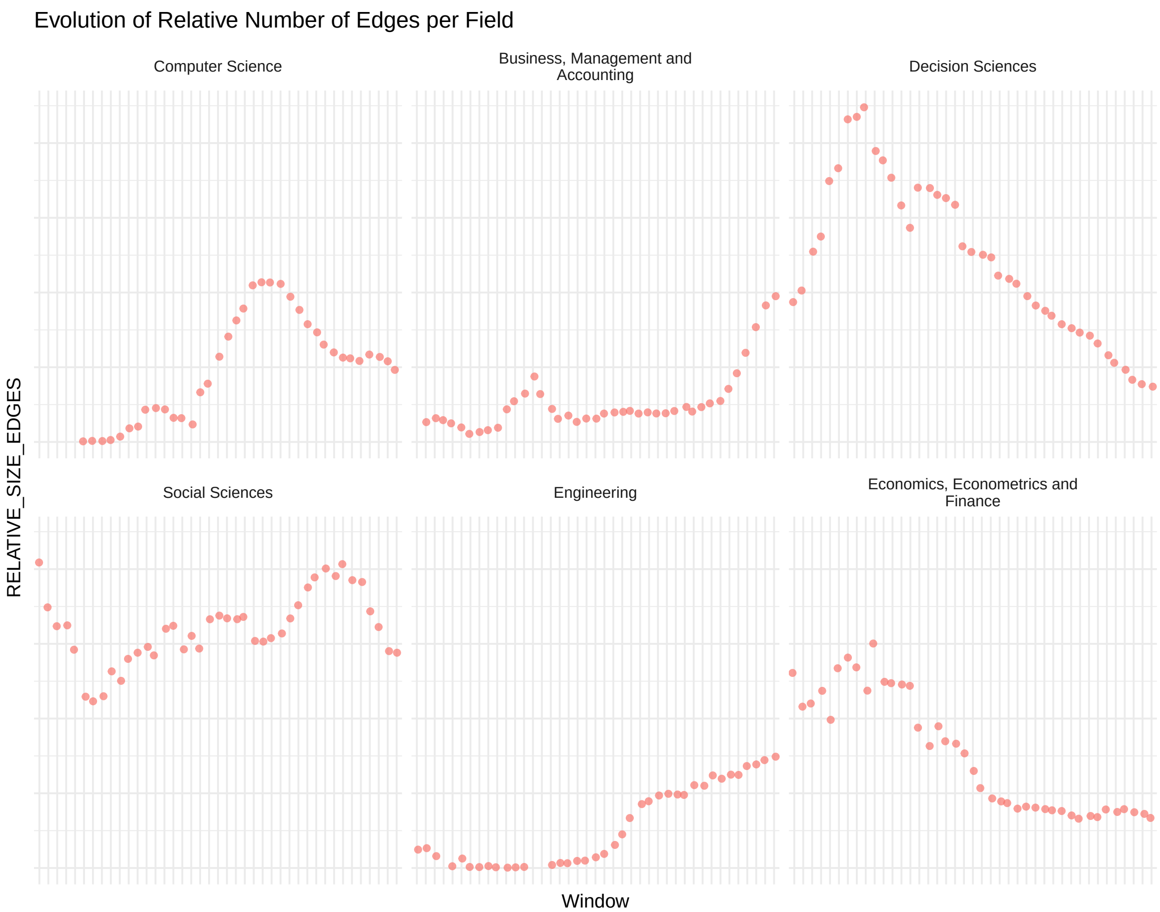
where:

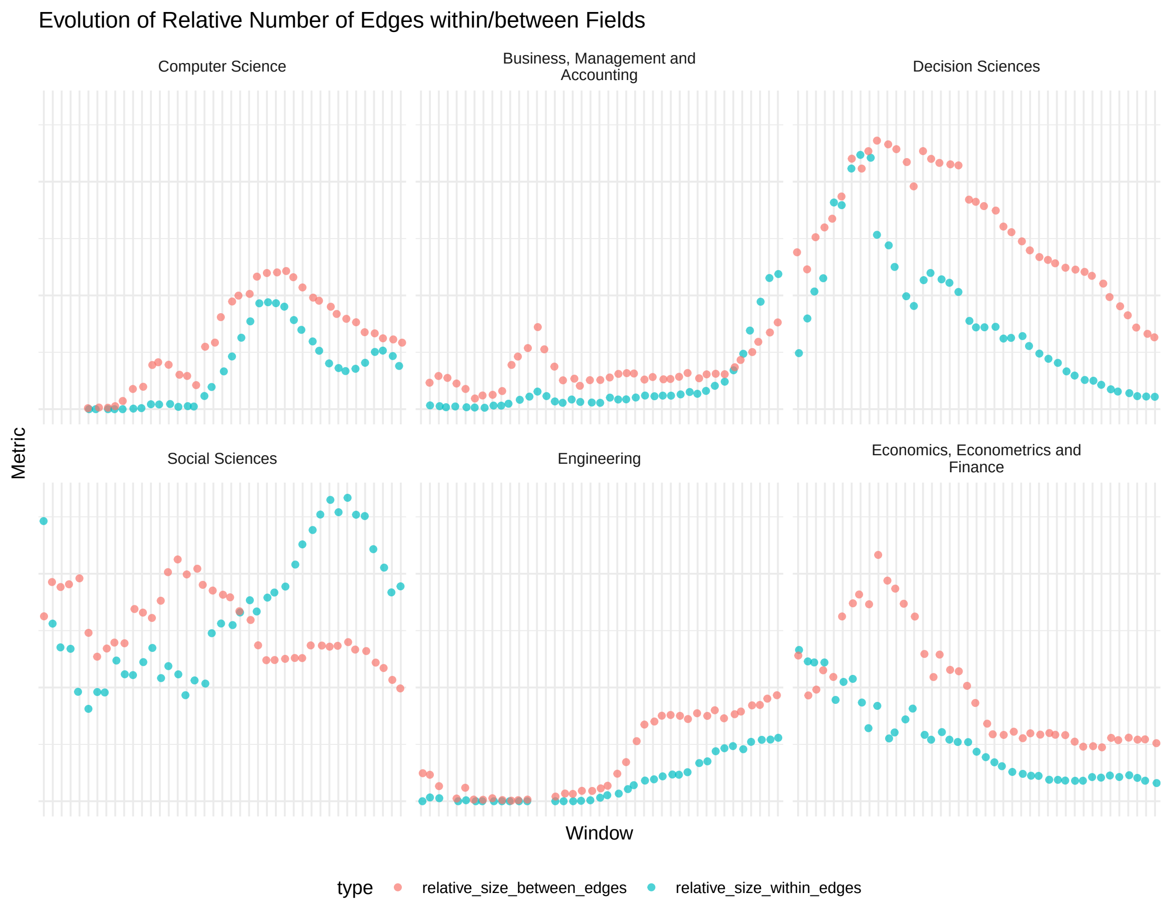
- $RelEdgesWithin_{t,f}$ measures the relative number of edges between article nodes of the same field for field f in time window t
- $N(E_{within})_{t,f}$ measures the total number of edges between article nodes of the same field for field f in time window t. It is therefore defined as $N(E_{within})_{t,f} = \sum_{e=(v_i,v_j)} 1_{field\{(v_i)=f\}} 1_{\{field(v_j)=f\}}$, where $1_{field(v_k)=f}$ is an indicator function, taking the value one if article node k belongs to field f.
- Relative Number of Between-Field Edges:

$$RelEdgesBetween_{t,f} = \frac{N(E_{Between})_{t,f}}{\sum_f^F N(E)_{t,f}} \tag{4}$$

where:

- $RelEdgesBetween_{t,f}$ measures the relative number of edges between article nodes of different fields for field f in time window t
- $N(E_{Between})_{t,f}$ measures the total number of edges between article nodes of different fields for field f in time window t. It's defined as $N(E_{Between})_{t,f} = \sum_{e=(v_i,v_j)} \left(1_{field\{(v_i=f)\}} 1_{field\{(v_j \neq f)\}} + 1_{field\{(v_i \neq f)\}} 1_{field\{(v_j=f)\}}\right)$.


Evolution of Relative Number of Edges per Field
Computer Science
Business, Management and Accounting
Decision Sciences
Social Sciences
Engineering
Economics, Econometrics and Finance
RELATIVE_SIZE_EDGES
Window



Evolution of Relative Number of Edges within/between Fields
Computer Science
Business, Management and Accounting
Decision Sciences
Social Sciences
Engineering
Economics, Econometrics and Finance
Metric
Window
type
relative_size_between_edges
relative_size_within_edges

2.3 Internal and External Edge Ratio

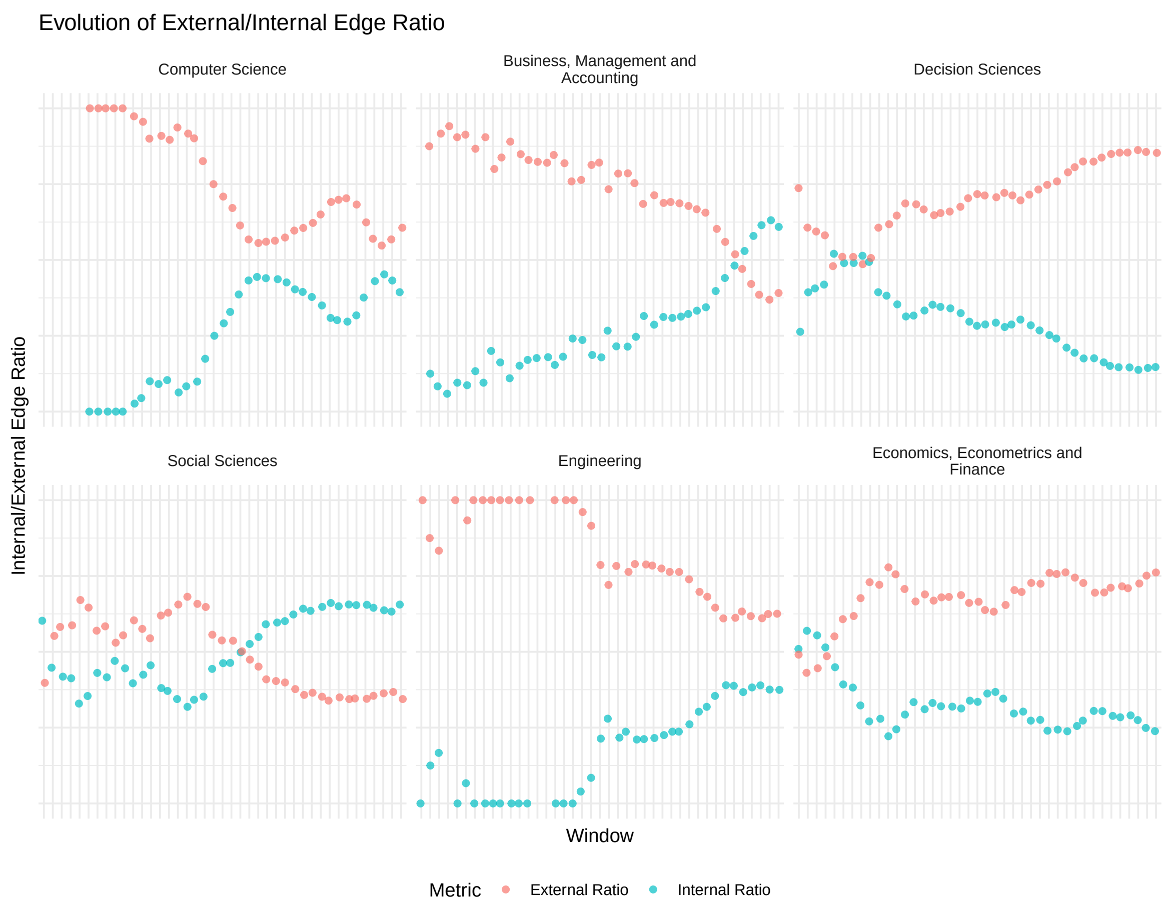
- Internal Edge Ratio:

$$IR_{t,f} = \frac{N(E_{within})_{t,f}}{N(E)_{t,f}} \tag{5}$$

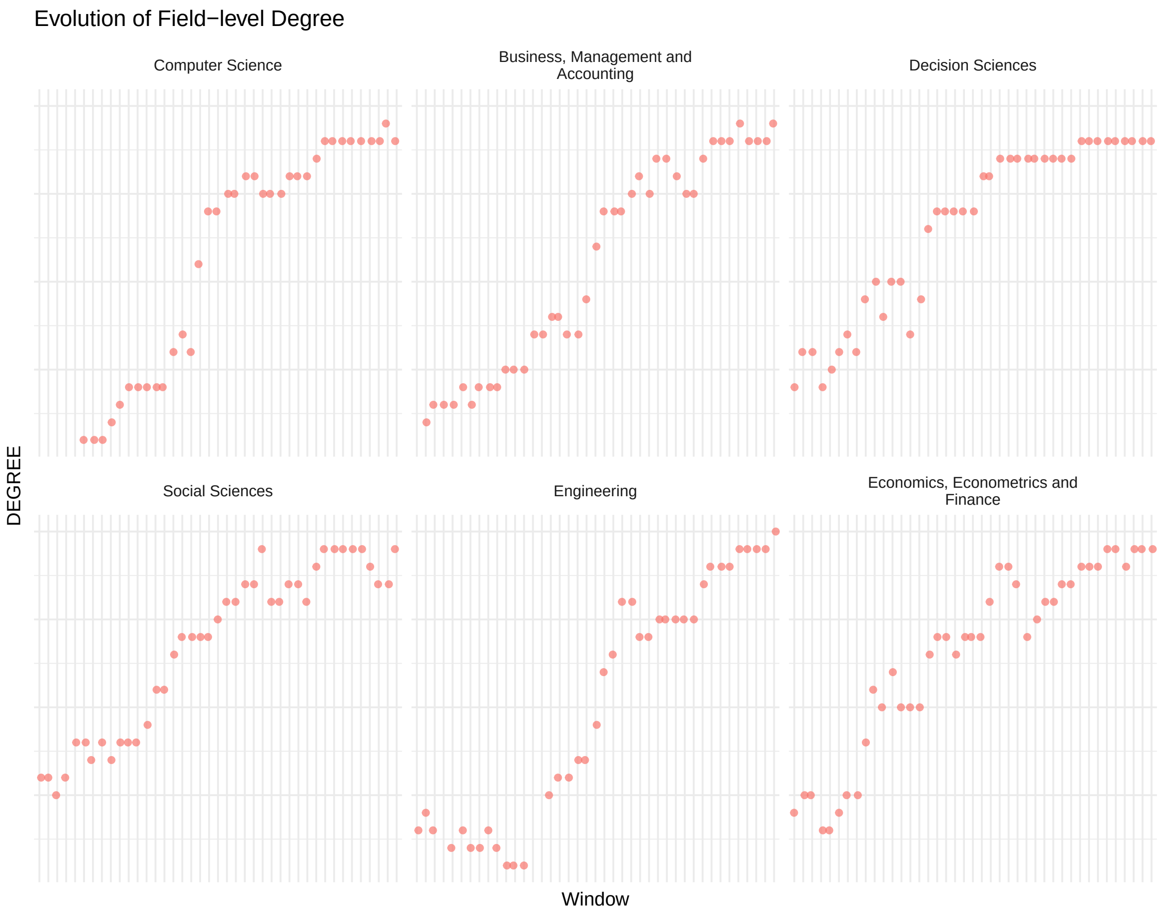
- External Edge Ratio:

$$ER_{t,f} = \frac{N(E_{between})_{t,f}}{N(E)_{t,f}} \tag{6}$$

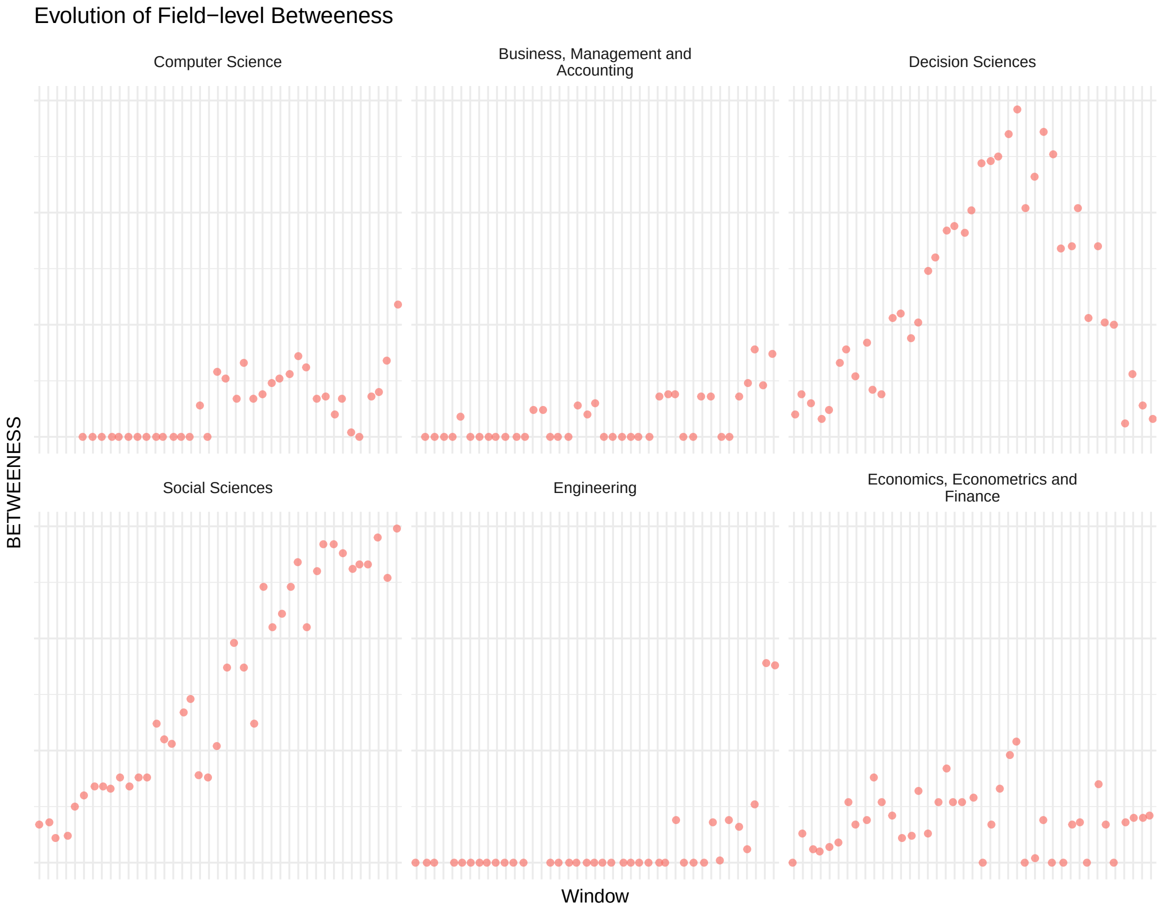
with $N(E_{within})_{t,f}$, $N(E_{between})_{t,f}$ and N(E)_{t,f} defined as before. The internal and external ratios therefore measure the ratio of edges within (between) fields relative to the total number of edges of a field. Edges are derived from article-level nodes.



2.4 Network Centrality Metrics (Degree, Closeness and Betweennes)



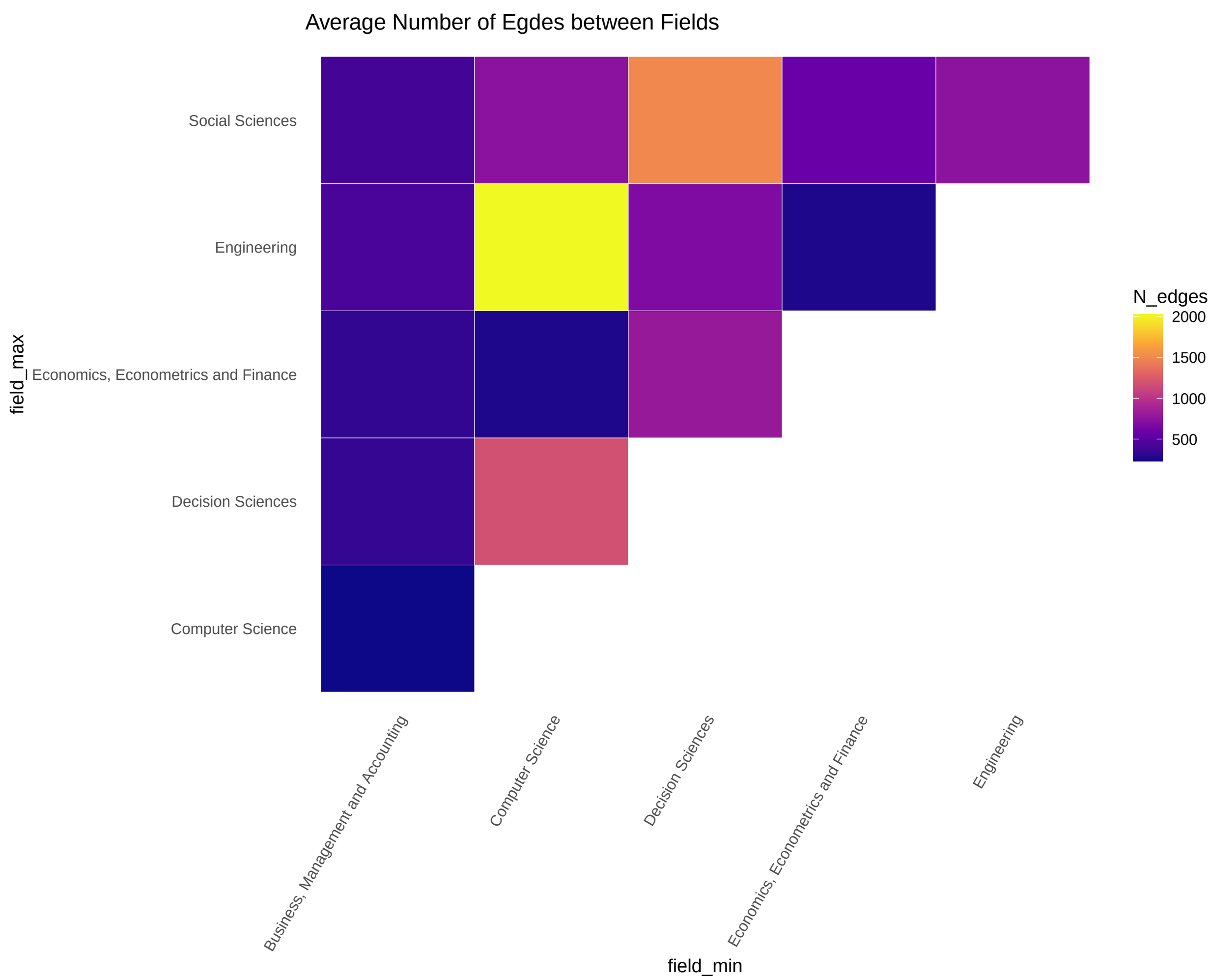

Evolution of Field−level Closeness
Computer Science
Business, Management and Accounting
Decision Sciences
Social Sciences
Engineering
Economics, Econometrics and Finance
CLOSENESS
Window

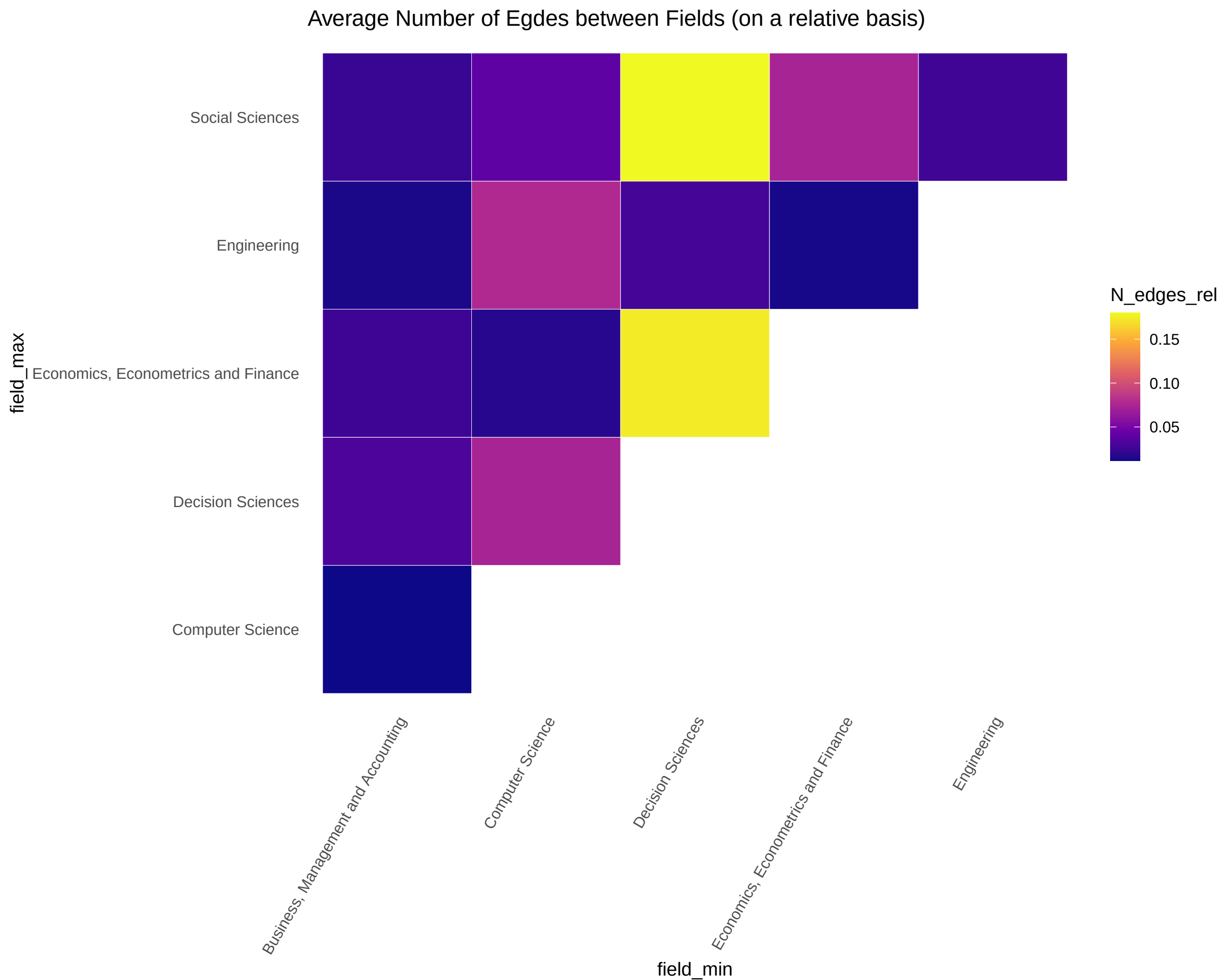


Evolution of Field−level Betweeness
Computer Science
Business, Management and Accounting
Decision Sciences
Social Sciences
Engineering
Economics, Econometrics and Finance
BETWEENESS
Window

3. Field-Field Connectivity

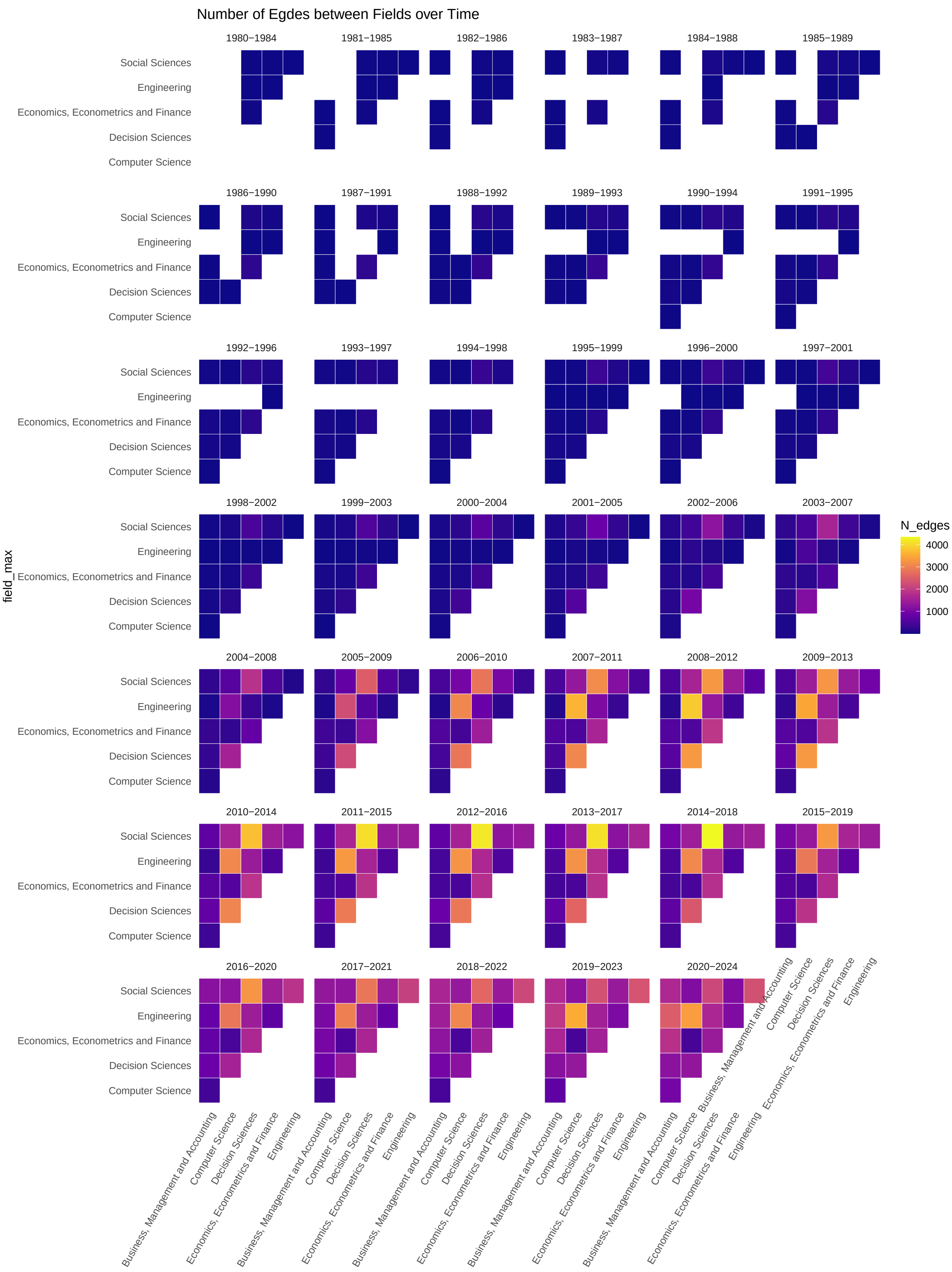
3.1 Average Connectivity

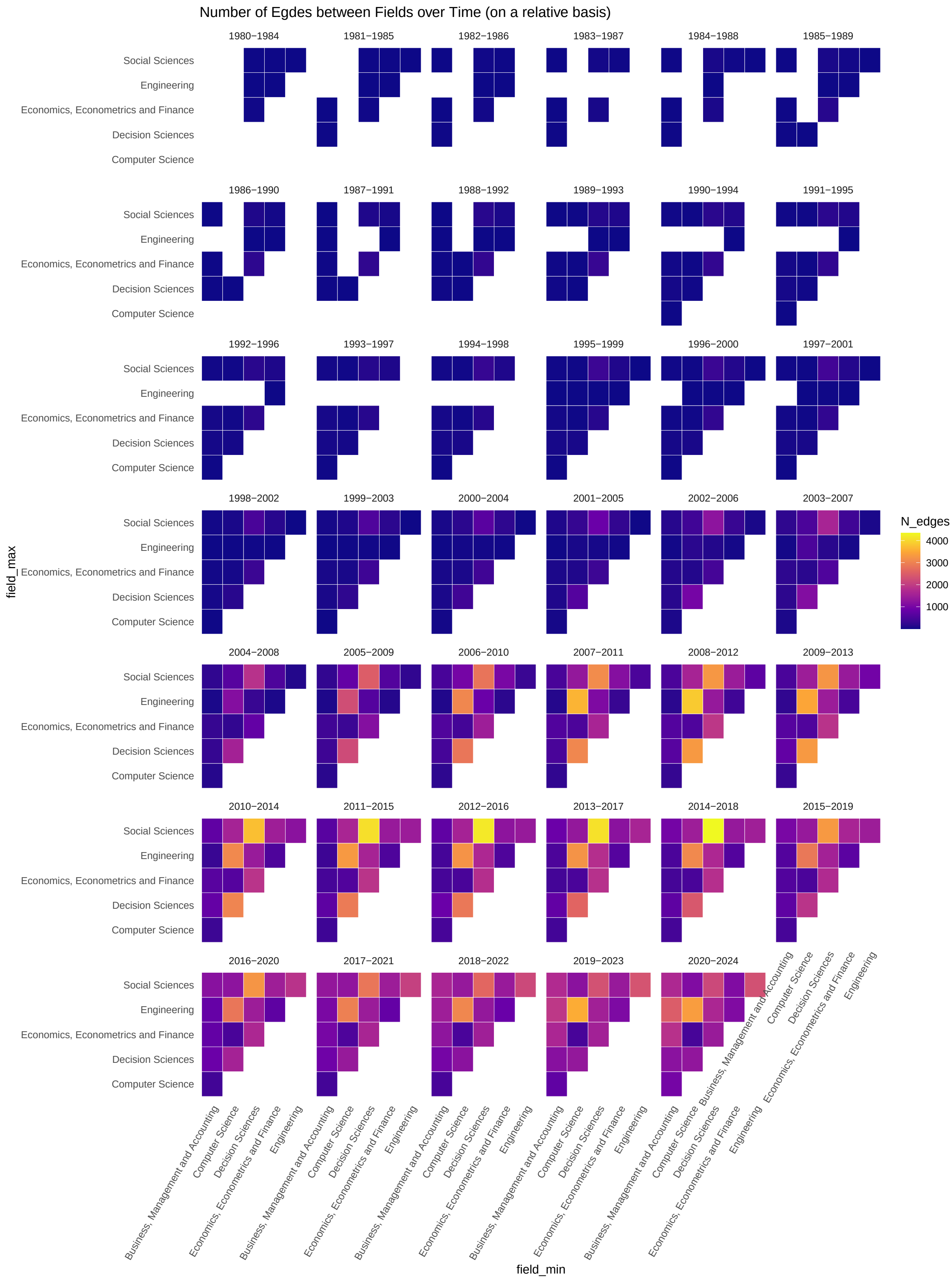



Average Number of Egdes between Fields (on a relative basis)
Social Sciences
Engineering
Economics, Econometrics and Finance
Decision Sciences
Computer Science
field_max
Business, Management and Accounting
Computer Science
Decision Sciences
Economics, Econometrics and Finance
Engineering
field_min
N_edges_rel
0.15
0.10
0.05

3.2 Connectivity over time

3.2.1 Heatmaps




Number of Egdes between Fields over Time (on a relative basis)
1980−1984
1981−1985
1982−1986
1983−1987
1984−1988
1985−1989
1986−1990
1987−1991
1988−1992
1989−1993
1990−1994
1991−1995
1992−1996
1993−1997
1994−1998
1995−1999
1996−2000
1997−2001
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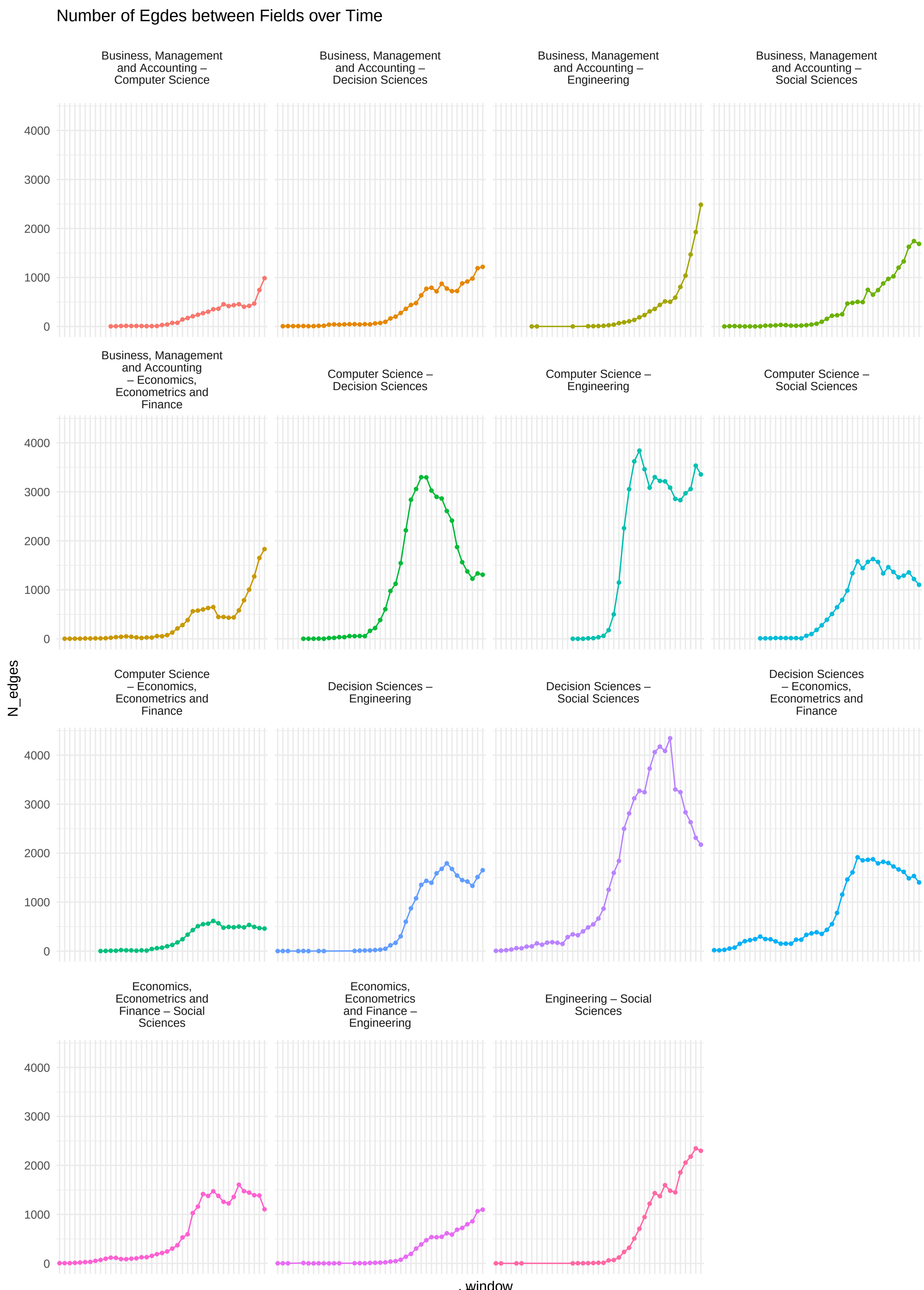
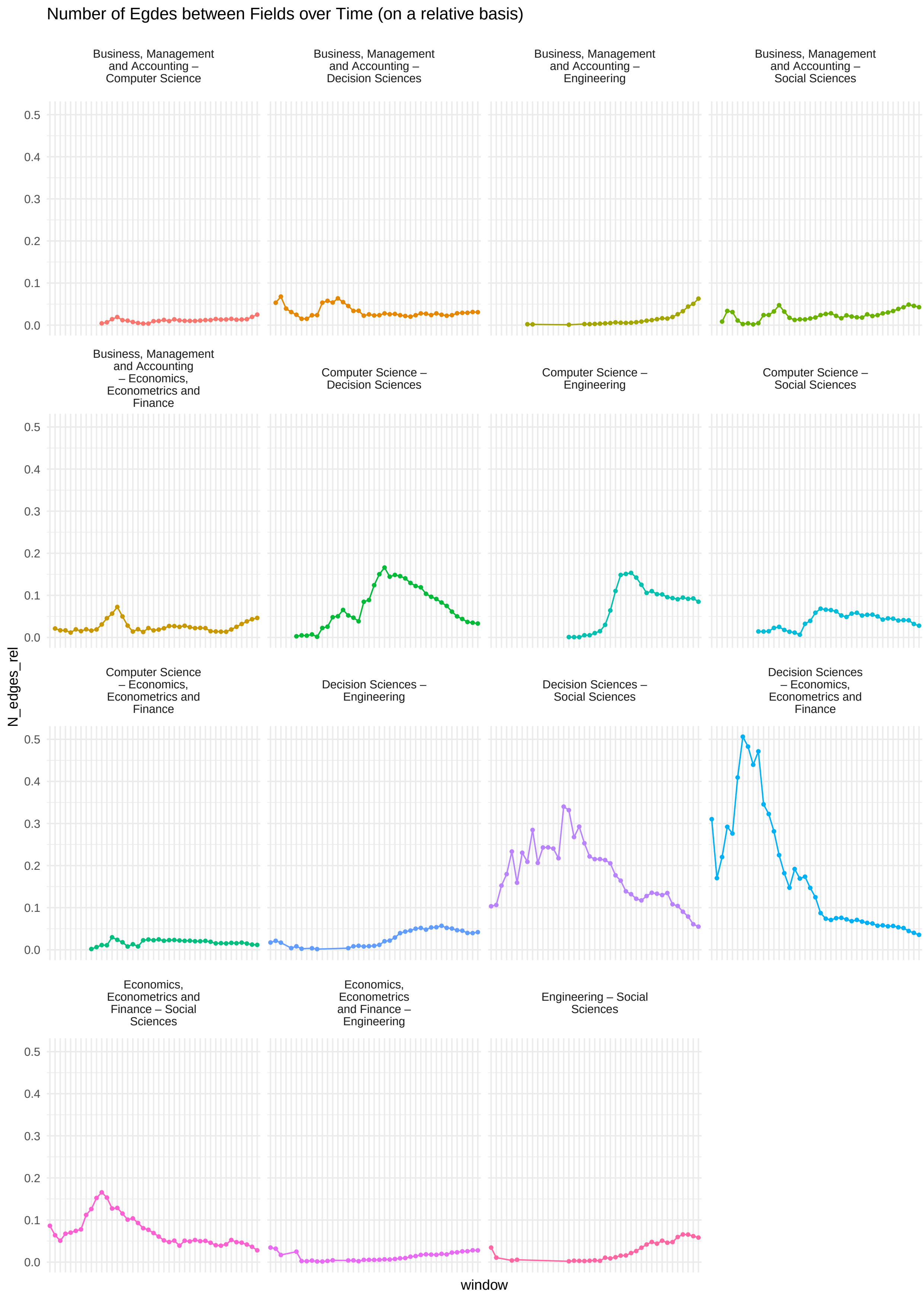
2007−2011
2008−2012
2009−2013
2010−2014
2011−2015
2012−2016
2013−2017
2014−2018
2015−2019
2016−2020
2017−2021
2018−2022
2019−2023
2020−2024
Social Sciences
Engineering
Economics, Econometrics and Finance
Decision Sciences
Computer Science
field_max
Business, Management and Accounting
Computer Science
Decision Sciences
Economics, Econometrics and Finance
Engineering
field_min
N_edges
4000
3000
2000
1000

3.2.2 Time Series




Number of Egdes between Fields over Time (on a relative basis)
Business, Management and Accounting – Computer Science
Business, Management and Accounting – Decision Sciences
Business, Management and Accounting – Engineering
Business, Management and Accounting – Social Sciences
Business, Management and Accounting – Economics, Econometrics and Finance
Computer Science – Decision Sciences
Computer Science – Engineering
Computer Science – Social Sciences
Computer Science – Economics, Econometrics and Finance
Decision Sciences – Engineering
Decision Sciences – Social Sciences
Decision Sciences – Economics, Econometrics and Finance
Economics, Econometrics and Finance – Social Sciences
Economics, Econometrics and Finance – Engineering
Engineering – Social Sciences
N_edges_rel
window
0.0
0.1
0.2
0.3
0.4
0.5

4. Citation Flows

4.1 Internal Coverage

Table 1: Number of References with Meta Data

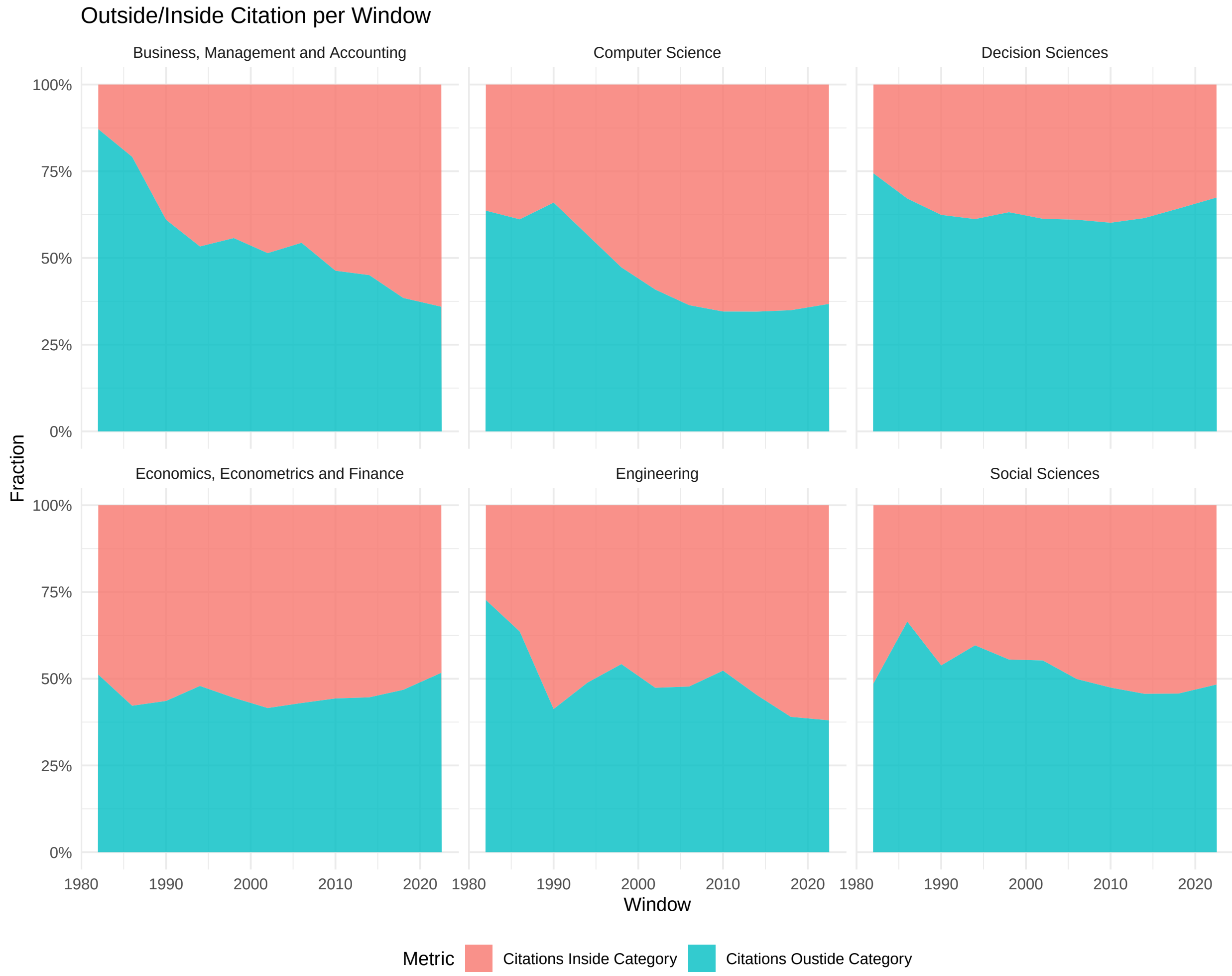
Field	Total Number of References	Relative Number of References (in %)
Agricultural and Biological Sciences	15878	1.45
Arts and Humanities	6257	0.57
Biochemistry, Genetics and Molecular Biology	15103	1.38
Business, Management and Accounting	122986	11.22
Chemical Engineering	101	0.01
Chemistry	587	0.05
Computer Science	172746	15.76
Decision Sciences	107728	9.83
Dentistry	24	0.00
Earth and Planetary Sciences	2806	0.26
Economics, Econometrics and Finance	134496	12.27
Energy	5772	0.53
Engineering	153172	13.98
Environmental Science	29274	2.67
Health Professions	4473	0.41
Immunology and Microbiology	867	0.08
Materials Science	777	0.07
Mathematics	17842	1.63
Medicine	10669	0.97
Neuroscience	5129	0.47
Nursing	172	0.02
Pharmacology, Toxicology and Pharmaceutics	158	0.01
Physics and Astronomy	20785	1.90
Psychology	14310	1.31
Social Sciences	146557	13.37
Veterinary	142	0.01
NA	106959	9.76

4.2 Outside/Inside Citations

4.2.1 Per Year

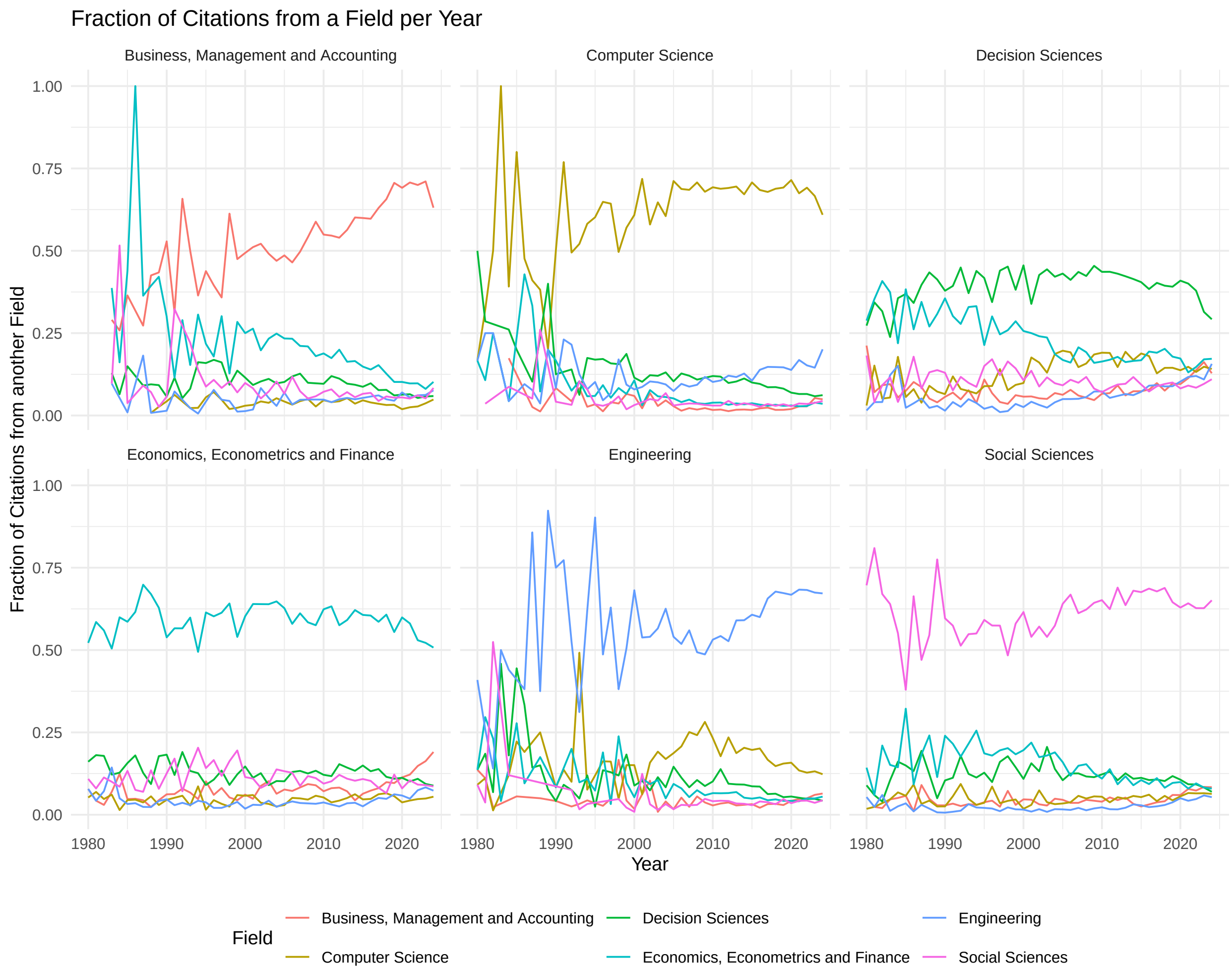

Outside/Inside Reference per Year
Business, Management and Accounting
Computer Science
Decision Sciences
Economics, Econometrics and Finance
Engineering
Social Sciences
Fraction
100%
75%
50%
25%
0%
1980
1990
2000
2010
2020
Year
Metric
Reference Inside Category
Reference Oustide Category

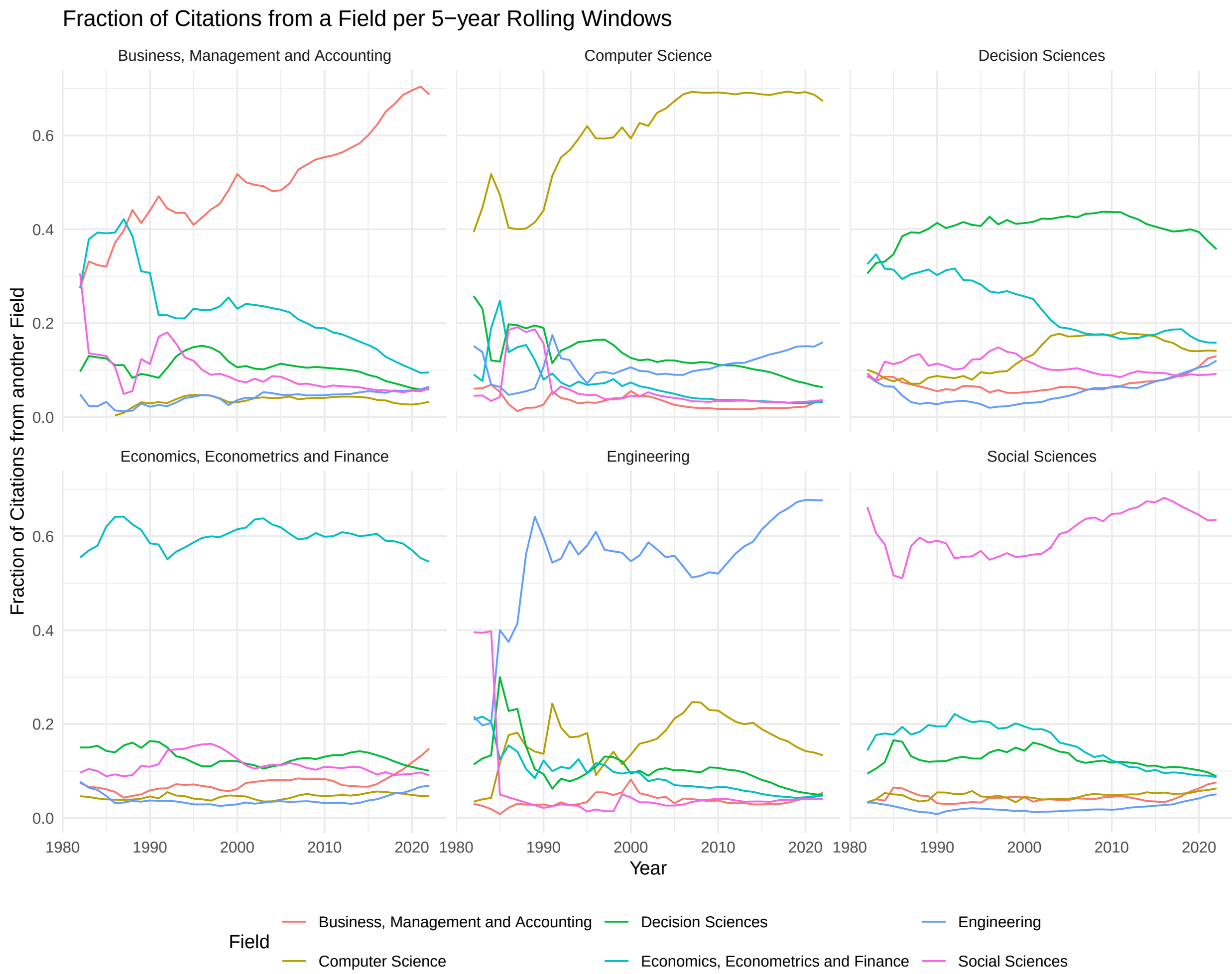
4.2.2 Per Window (non-overlapping)



Outside/Inside Reference per Window
Business, Management and Accounting
Computer Science
Decision Sciences
Economics, Econometrics and Finance
Engineering
Social Sciences
Fraction
100%
75%
50%
25%
0%
1980
1990
2000
2010
2020
Window
Metric
Reference Inside Category
Reference Oustide Category

4.3 Citation Flows over Time




Fraction of Citations from a Field per 5−year Rolling Windows
Business, Management and Accounting
Computer Science
Decision Sciences
Economics, Econometrics and Finance
Engineering
Social Sciences
Fraction of Citations from another Field
0.0
0.2
0.4
0.6
1980
1990
2000
2010
2020
Year
Field
Business, Management and Accounting
Computer Science
Decision Sciences
Economics, Econometrics and Finance
Engineering
Social Sciences

4.4 Alluvial Graph of Flows